Expressing controlled values: Historical background

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Darwin Core Maintenance Group Meeting

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Development of the Standards Documentation Specification

- VOCAB Task Group:
  - chartered in May 2015, I was convener
  - tasked with developing the SDS and the VMS
  - based work on existing TDWG practice and best practices of (aspirational) peer organizations (W3C, IETF, and DCMI)
  - meeting notes: https://github.com/tdwg/vocab/tree/master/meeting-notes
  - completed drafts in July 2016, SDS ratified April 2017
- Precedents for format of human-readable docs and RDF from Darwin Core (modelled on Dublin Core)
- Controlled vocabularies recommended, but none existed within TDWG.

What should a “controlled vocabulary” be?

- A list of controlled string values?
- An ontology?
- A knowledge organization system (KOS)?
- Should it have features of existing TDWG vocabularies (DwC and AC)?

Examination of ISO 25964-1 and -2

- Infuriatingly under a paywall, but see notes at https://github.com/tdwg/vocab/blob/master/iso25964.md
- Contained definitions of ”controlled vocabulary” and described differences between an ontology and thesaurus approach (Section 21 of ISO 25964-2)
- Described the use of thesauri in directing users to the correct concept through definitions, hierarchies, and labels, with a focus on information retrieval rather than describing reality.

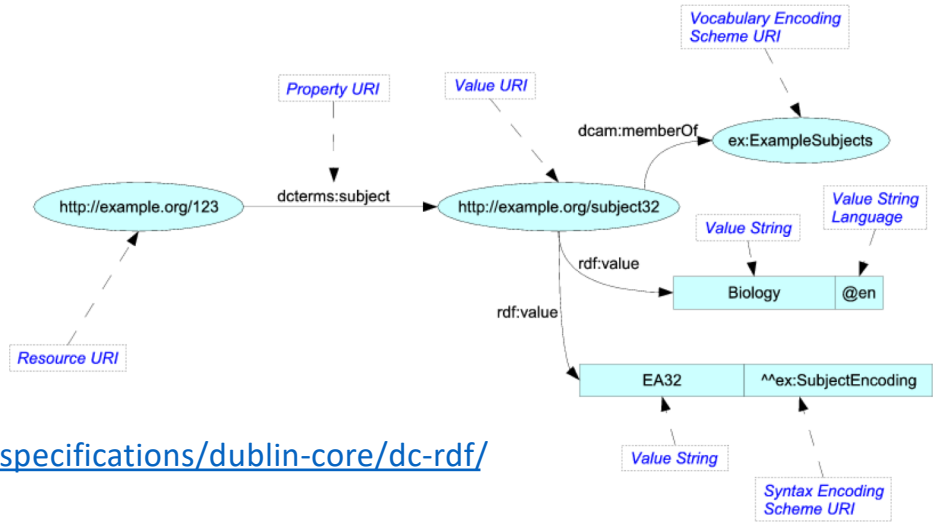
SKOS as a model for controlled vocabularies

- Simple Knowledge Organization System (SKOS) clearly developed with the intention of realizing ISO 25964 in the context of the Web and Linked Data.
- Support for preferred labels, multilingual labels, definitions, and hierarchies of concepts – all features of importance in the context of TDWG
- Implementable in machine-readable RDF (important given that the SDS was intended to specify machine-readable serializations of vocabularies)
- VOCAB TG decided to consider TDWG controlled vocabularies to be SKOS concept schemes

Key deficiency of SKOS

- Primary identification of concepts was by IRIs, not strings.
- Different preferred label supported for each language
  - China = English
  - Zhongguo = transliterated Chinese
  - 中华人民共和国 = Chinese characters
- No mechanism or property for for specifying a single preferred controlled string value

Dublin Core RDF Conceptual Model



https://www.dublincore.org/specifications/dublin-core/dc-rdf/

- As Darwin Core was based on Dublin Core, the VOCAB group looked to DCMI for a model. The DC Conceptual Model used rdf:value as the property to indicate the value of a concept.
- See https://www.w3.org/TR/2004/REC-rdf-primer-20040210/#rdfvalue for examples of use of rdf:value
- The VOCAB TG chose rdf:value as the property to indicate the language-independent controlled value that should be used in text systems (e.g. spreadsheets)

Controlled vocabulary-specific terms in the SDS

- Section 4.5.4 of the SDS describes how SKOS should be used to define the relationship between controlled vocabulary concept terms and other resources such as labels, definitions, and other concepts.
- It is careful to distinguish between preferred labels and controlled text string values that can be used to identify the concept in lieu of an IRI.
- “However, in the context of information transfer, there might be cases where it is more desirable to represent the controlled vocabulary term by means of a unique text string. To facilitate such use, each controlled vocabulary term MUST be assigned a string that is unique within that controlled vocabulary. That string SHOULD be designated by making it a plain literal value (without language tag) of an rdf:value property of the term.”

Creation of the first ratified TDWG vocabularies

- Although the SDS was ratified in 2017, the first controlled vocabularies were not created until 2019, with ratification in October 2020: establishmentMeans, pathway, and degreeOfEstablishment
- The SDS is silent about the form of IRIs and controlled value strings. At the TAG meeting in Leiden in 2019, there was discussion about controlled vocabulary IRI form (opaque or not?) and controlled strings. The meeting ended without a conclusion and there were no regularly scheduled TAG meetings.
- Goal: make clear the distinction between labels in many languages and the single controlled value string.

IRI and label patterns

- Darwin Core:
  - Label: “Coordinate Uncertainty in Meters”
  - IRI: http://rs.tdwg.org/dwc/terms/coordinateUncertaintyInMeters
  - CURIE: dwc:coordinateUncertaintyInMeters

  (confusion between English label, CURIE, and term local name)

- Controlled vocabulary
  - English Label: “introduced: assisted colonization”
  - Spanish label: ”introducido: colonización asistida”
  - Russian label: “интродуцирован: преднамеренная интродукция”
  - IRI: http://rs.tdwg.org/dwcem/values/e004
  - CURIE: dwcem:e004
  - Controlled value string: “introducedAssistedColonisation”

  (established precedent of lowerCamelCase)

When not to follow the precedent: Existing example values and stability

- Exception 1: In some cases, there are existing standard strings (e.g. AC's format controlled vocabulary, http://rs.tdwg.org/ac/doc/format/ which uses standard media types and file extensions, e.g., "image/x-canon-crw" and "crw"
- Exception 2: In 2020, AC's variant controlled vocabulary formalized variant types that had been given as examples for ac:variant since 2013. It would have been disruptive to stability to change the example strings e.g., "Lower Quality".
- Otherwise, the lowerCamelCase has been used in all existing ratified controlled vocabularies in AC and DwC where values were modelled as SKOS concepts (content description, subjectOrientation, subjectPart, Establishment Means, Degree of Establishment, Pathway, and Taxon Completeness Reported)

Darwin Core term examples and lowerCamelCase use

- Cases where “best practice is to use a controlled vocabulary” but no vocabulary outside TDWG is recommended.
- lowerCamelCase used: dwc:agentType, dwc:assayType, dwc:assertionType, dwc:eventType, dwc:lifeStage, dwc:organismInteractionType, dwc:protocolType, dwc:referenceType, dwc:sampledSubstrateCategory, dwc:sex, dwc:vitality, eco:compliationSourceTypes, eco:compilationTypes, eco:eventDurationUnit, eco:excludedDegreeOfEstablishmentScope, eco:excludedGrowthFormScope, eco:excludedHabitatScope, eco:excludedLifeStageScope, eco:includeOrExclude, eco:inventoryTypes, eco:materialSampleTypes, eco:protocolNames, eco:samplingEffortUnit, eco:surveyTargetType (with one error), eco:targetDegreeOfEstablishmentScope, eco:targetGrowthFormScope, eco:targetHabitatScope, eco:targetLifeStageScope, eco:taxonCompletenessReported (29 terms)
- Pre-CV terms not using lowerCamelCase: dwc:disposition (2007), dwc:georeferenceVerificationStatus (2007), dwc:measurementType (2008), dwc:organismScope (2014), dwc:relationshipOfResource (2009), dwc:reproductiveCondition (2009), dwc:taxonomicStatus (2009), dwc:verbatimCoordinateSystem (2009) (8 terms)
- Post-CV terms not using lowerCamelCase: dwc:caste (2023), dwc:discipline (2025), dwc:verticalDatum (2021 but codes), eco:surveyTargetValue (2026), dwc:sampledSubstrateLayer (2026) (5 terms)
- dwc:basisOfRecord is a special case where values are classes and not concepts, similar to ac:subtype. In these cases, the local name of the term IRI is used by convention.